

Pavel Tokmakov

Senior Research Scientist, Toyota Research Institute

World models · Robot learning · Video understanding

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Google Scholar 

pvtokmakov.github.io/home 

West Hollywood, CA, USA 

RESEARCH EXPERIENCE

- 2020 – now **Senior Research Scientist** Los Altos, USA
[Toyota Research Institute](#)
Leading research on world models for embodied AI. Conducted a large-scale study of where scaling falls short for ego-centric world models trained on human video. Developed some of the first methods across several directions: video generators as robot policies, generative models for image and video understanding, interpreting video transformer representations, and tracking objects through occlusion.
- 2018 – 2020 **Postdoctoral Researcher** Pittsburgh, USA
[CMU Robotics Institute](#) · with Prof. Martial Hebert and Prof. Deva Ramanan
Co-directed research projects and conducted independent research on video understanding and learning with limited supervision. Developed compositional representations for few-shot recognition and introduced TAO, the first large-scale benchmark for open-world tracking.
- 2014 – 2018 **PhD Student** Grenoble, France
[Inria THOTH team](#) · with Dr. Cordelia Schmid and Dr. Karteek Alahari
Studied the role of motion in object recognition. Developed the first learning-based methods for motion understanding in video, and showed how motion can supervise object recognition.

EDUCATION

- 2014 – 2018 **PhD in Computer Science** France
[University of Grenoble](#)
Thesis: *Weakly-Supervised Learning for Object Recognition*
- 2012 – 2014 **MSc in Computer Science** Germany
[University of Bonn](#)
Emphasis on Intelligent Systems · GPA 4.0

SELECTED PUBLICATIONS

* Equal contribution

- Preprint**
2026 **What 30,000 Hours of Ego-centric Video Does Not Teach**
Jiahua Dong, Anurag Bagchi, Yash Jangir, Zubair Irshad, Sergey Zakharov, Martial Hebert, Homanga Bharadhwaj, Yu-Xiong Wang, Vitor Guizilini, [Pavel Tokmakov](#)
- Preprint**
2026 **GPWM: Generative Particle World Models for 3D Prediction and Planning**
Robert Ren, Matthew Bronars, He Zhu, Sergey Zakharov, [Pavel Tokmakov](#), Katerina Fragkiadaki
- ECCV**
2026 **Walk Through Paintings: Ego-centric World Models from Internet Priors**
Anurag Bagchi, Zhipeng Bao, Homanga Bharadhwaj, Yu-Xiong Wang, [Pavel Tokmakov*](#), Martial Hebert*
- ICLR**
2026 **Capturing Visual Environment Structure Correlates with Control Performance**
Jiahua Dong, Yunze Man, [Pavel Tokmakov*](#), Yu-Xiong Wang*
- Sci. Rob.**
2026 **A Careful Examination of Large Behavior Models for Multitask Dexterous Manipulation**
TRI LBM Team
- arXiv**
2025 **Video Generators Are Robust Robot Policies**
Junbang Liang, [Pavel Tokmakov](#), Ruoshi Liu, Ege Ozguroglu, Sruthi Sudhakar, Paarth Shah, Rares Ambrus, Carl Vondrick
- ICCV**
2025 **ReferEverything: Towards Segmenting Everything We Can Speak of in Videos**
Anurag Bagchi, Zhipeng Bao, Yu-Xiong Wang, [Pavel Tokmakov*](#), Martial Hebert*
- ICML**
2025 **Analyzing Question Complexity in Video QA via Visual Program Generation**
Cristobal Eyzaguirre, Igor Vasiljevic, Achal Dave, Jiajun Wu, Thomas Kollar, Juan Carlos Niebles, [Pavel Tokmakov](#)

ECCV	Generative Camera Dolly: Extreme Monocular Dynamic Novel View Synthesis ORAL
2024	Basile Van Hoorick, Rundi Wu, Ege Ozguroglu, Kyle Sargent, Ruoshi Liu, <u>Pavel Tokmakov</u> , Achal Dave, Changxi Zheng, Carl Vondrick
CoRL	Dreamitate: Real-world Visuomotor Policy Learning via Video Generation
2024	Junbang Liang, Ruoshi Liu, Ege Ozguroglu, Sruthi Sudhakar, Achal Dave, <u>Pavel Tokmakov</u> , Shuran Song, Carl Vondrick
CVPR	Understanding Video Transformers via Universal Concept Discovery SPOTLIGHT
2024	Matthew Kowal, Achal Dave, Rares Ambrus, Adrien Gaidon, Konstantinos Derpanis, <u>Pavel Tokmakov</u>
CVPR	pix2gestalt: Amodal Segmentation by Synthesizing Wholes SPOTLIGHT
2024	Ege Ozguroglu, Ruoshi Liu, Didac Suris, Dian Chen, Achal Dave, <u>Pavel Tokmakov</u> , Carl Vondrick
ICCV	Zero-1-to-3: Zero-shot One Image to 3D Object
2023	Ruoshi Liu, Rundi Wu, Basile Van Hoorick, <u>Pavel Tokmakov</u> , Sergey Zakharov, Carl Vondrick
CVPR	Breaking the “Object” in Video Object Segmentation
2023	<u>Pavel Tokmakov</u> , Jie Li, Adrien Gaidon
ICML	Object Permanence Emerges in a Random Walk along Memory
2022	<u>Pavel Tokmakov</u> , Allan Jabri, Jie Li, Adrien Gaidon
CVPR	Discovering Objects that Can Move
2022	Zhipeng Bao*, <u>Pavel Tokmakov</u> *, Allan Jabri, Yu-Xiong Wang, Adrien Gaidon, Martial Hebert
ICCV	Learning to Track with Object Permanence
2021	<u>Pavel Tokmakov</u> , Jie Li, Wolfram Burgard, Adrien Gaidon
ECCV	TAO: A Large-Scale Benchmark for Tracking Any Object SPOTLIGHT
2020	Achal Dave, Tarasha Khurana, <u>Pavel Tokmakov</u> , Cordelia Schmid, Deva Ramanan
ICCV	Learning Compositional Representations for Few-Shot Recognition
2019	<u>Pavel Tokmakov</u> , Yu-Xiong Wang, Martial Hebert
CVPR	A Structured Model For Action Detection
2019	Yubo Zhang, <u>Pavel Tokmakov</u> , Cordelia Schmid, Martial Hebert
ICCV	Learning Video Object Segmentation with Visual Memory ORAL
2017	<u>Pavel Tokmakov</u> , Karteek Alahari, Cordelia Schmid
CVPR	Learning Motion Patterns in Videos
2017	<u>Pavel Tokmakov</u> , Karteek Alahari, Cordelia Schmid

HONORS & AWARDS

2026	PAMI Mark Everingham Prize · awarded to the Visual Object Tracking (VOT) Challenge Series team and collaborators, for the series of workshops and challenges that have shaped the modern synthesis of visual object tracking
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WORKSHOPS & ORGANIZATION

Each workshop ran with an accompanying challenge. Responsibilities: proposals, speaker recruitment, program, panel moderation.

2024 – 2026	VOTS Workshop Series · ICCV / ECCV · co-organizer · video segmentation
2023	5th LSVOS Workshop · ICCV, Paris · lead organizer · video segmentation
2023	2nd TAO Workshop · CVPR, Vancouver · co-organizer · open-world tracking
2021	RVSU Workshop · CVPR · co-organizer · robust video scene understanding
2020	TAO Workshop · ECCV · lead organizer · tracking any object, new benchmark
2018 – 2020	IARPA DIVA Project · CMU · coordinated the CMU team on spatio-temporal action localization

PROFESSIONAL SERVICE

2015 – now	Area Chair / Reviewer NeurIPS · ICLR · CVPR · ICCV · ECCV · ICML · CoRL
2019 – 2020	CMU MSCV Admission Committee Evaluating applications for the Master’s in Computer Vision program